

04) Linear Regression

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01) a) $\hat{y} = 6x + 10$

$x = 4$

$y = 6 \cdot 4 + 10 = 54 + 10$

$y = 64$

$[4, 64]$

$e_i = y_i - \hat{y}_i$

b) $x = 5$ $y = 6 \cdot 5 + 10 = 40$

$e_5 = 41 - 40$

$e_5 = 1$

$e^2 = 1$

c) $x = 1$ $y = 6 \cdot 1 + 10 = 16$

$e_1 = 19 - 16 = 3$

$e^2 = 9$

$x = 3$ $y = 6 \cdot 3 + 10 = 28$

$e_3 = 25 - 28 = -3$

$e^2 = 9$

$x = 7$ $y = 6 \cdot 7 + 10 = 52$

$e_7 = 51 - 52 = -1$

$e^2 = 1$

X	y	$\hat{y}$	e	e^2
1	19	16	3	9
3	25	28	-3	9
5	41	40	1	1
7	51	52	-1	1

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

$$\frac{(19-16)^2 + (25-28)^2 + (41-40)^2 + (51-52)^2}{4}$$

$$\frac{9 + 9 + 1 + 1}{4} = 5 \quad \underline{\underline{MSE = 5}}$$

$$RMSE = \sqrt{5} = \underline{\underline{2.23}}$$

$$SSE = 20$$

$$\bar{y} = \frac{19 + 25 + 41 + 51}{4} = 34$$

$$SST = \sum (y_i - \bar{y})^2$$

$$\begin{aligned} & \frac{-15}{(19-34)^2} + \frac{-9}{(25-34)^2} + \frac{7}{(41-34)^2} + \frac{17}{(51-34)^2} \\ & 225 + 81 + 49 + 289 = 644 \end{aligned}$$

$$\underline{\underline{SSE = 20}}$$

$$\underline{\underline{SST = 644}}$$

$$R^2 = 1 - \frac{SSE}{SST}$$

$$R^2 = 1 - \frac{20}{644} = 0.96894$$

$$\underline{\underline{R^2 = 0.969}}$$

line explains about

The result means that the ~~most~~ ~~in~~ ~~the~~ score ~~is~~ ~~about~~ 96.9%.